

SYLLABUS :-

Name of the course: Systems Programming Laboratory Proposed Course No.: CS29008
Type of Course: Core Laboratory Course for both BTech 4-Year Course and Dual Degree 5-Year Integrated BTech-MTech Course Semester: 4L-T-P: 0-0-3 Credit: 2
Syllabus: x [Lab 1]: Introduction to the course, writing makefiles, using gcc/clang (3 hours) x [Lab 2]: Introduction to code debugging, understanding code debugging with gdb (3 hours) x [Lab 3]: Code profiling using gprof, finding inefficiencies and performance bottlenecks in code, optimizing code performance (3 hours) x [Lab 4]: Memory profiling, debugging, and management using valgrind (3 hours) x [Lab 5]: Setting up and using version control systems (3 hours)x [Lab 6-7]: Unix-based commands for system management, file processing, pipes, regular expressions and text processing using grep, sed, and awk (6 hours) x [Lab 8-9]: Unix shells and shell scripting (6 hours) x [Lab 10-11]: Introduction to linking and loading programs, Binary file formats - Executable and Linkable File Formats (ELF), analyzing the hexdump of a binary file for analyzing program headers (6 hours) x [Lab 12-13] System configuration and management (6 hours)